

1-5: Convert the following binary numbers to decimal: 10.10001, 101110.01, 1110101.110, 1101101.11

$$10.10001 \rightarrow 1 \times 2^1 + 1 \times 2^{-1} + 1 \times 2^{-5} = 2.53125$$

$$101110.0101 \rightarrow 1 \times 2^5 + 1 \times 2^4 + 1 \times 2^3 + 1 \times 2^2 + 1 \times 2^{-2} + 1 \times 2^{-4} = 46.3125$$

$$1110101.110 \rightarrow 1 \times 2^6 + 1 \times 2^5 + 1 \times 2^4 + 1 \times 2^3 + 1 \times 2^2 + 1 \times 2^{-1} + 1 \times 2^{-2} = 117.75$$

$$1101101.111 \rightarrow 1 \times 2^6 + 1 \times 2^5 + 1 \times 2^4 + 1 \times 2^3 + 1 \times 2^2 + 1 \times 2^{-1} + 1 \times 2^{-2} + 1 \times 2^{-3} = 109.875 \square$$

1-6: Convert the following numbers from the given base to the bases indicated:
(a) decimal 225.225 to binary, octal, and hexadecimal

$$\begin{array}{r|l} 2 & 225 \\ \hline & 112 \\ 2 & 56 \\ 2 & 28 \\ 2 & 14 \\ 2 & 7 \\ 2 & 3 \\ 2 & 1 \\ & 0 \end{array}$$

$$0.225 \times 2 = 0.450$$

$$0.450 \times 2 = 0.90$$

$$0.90 \times 2 = 1.8$$

$$0.8 \times 2 = 1.6$$

$$0.6 \times 2 = 1.2$$

$$0.2 \times 2 = 0.4$$

$$0.4 \times 2 = 0.8$$

$$0.8 \times 2 = 1.6 \dots$$

$$(225.225)_2 = 11100001.0011001\dots$$

$$\begin{array}{r|l} 8 & 225 \\ \hline & 28 \\ 8 & 3 \\ & 0 \end{array}$$

$$0.225 \times 8 = 1.800$$

$$0.8 \times 8 = 6.4$$

$$0.4 \times 8 = 3.2$$

$$0.2 \times 8 = 1.6$$

$$0.6 \times 8 = 4.8$$

$$0.8 \times 8 = 6.4$$

$$(225.225)_8 = 341.16314\dots$$

$$\begin{array}{r|l} 16 & 225 \\ \hline & 14 \\ 16 & 0 \end{array} \quad \text{E}$$

$$0.225 \times 16 = 3.6$$

$$0.6 \times 16 = 9.6$$

$$0.6 \times 16 = 9.6$$

$$(225.225)_{16} = \text{E}1.399\dots$$

(b) binary 1101011.110 to decimal, octal and hexadecimal.